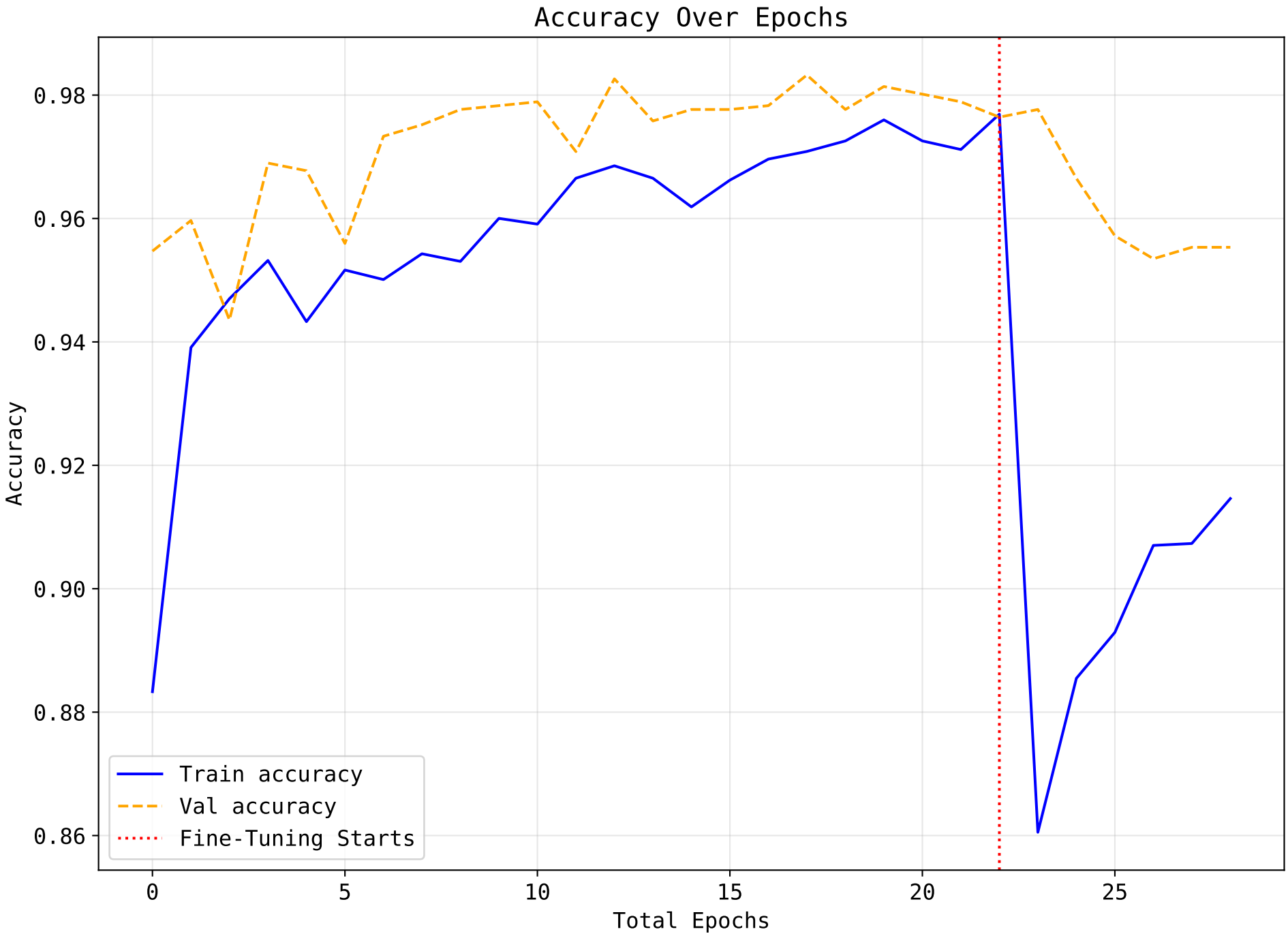
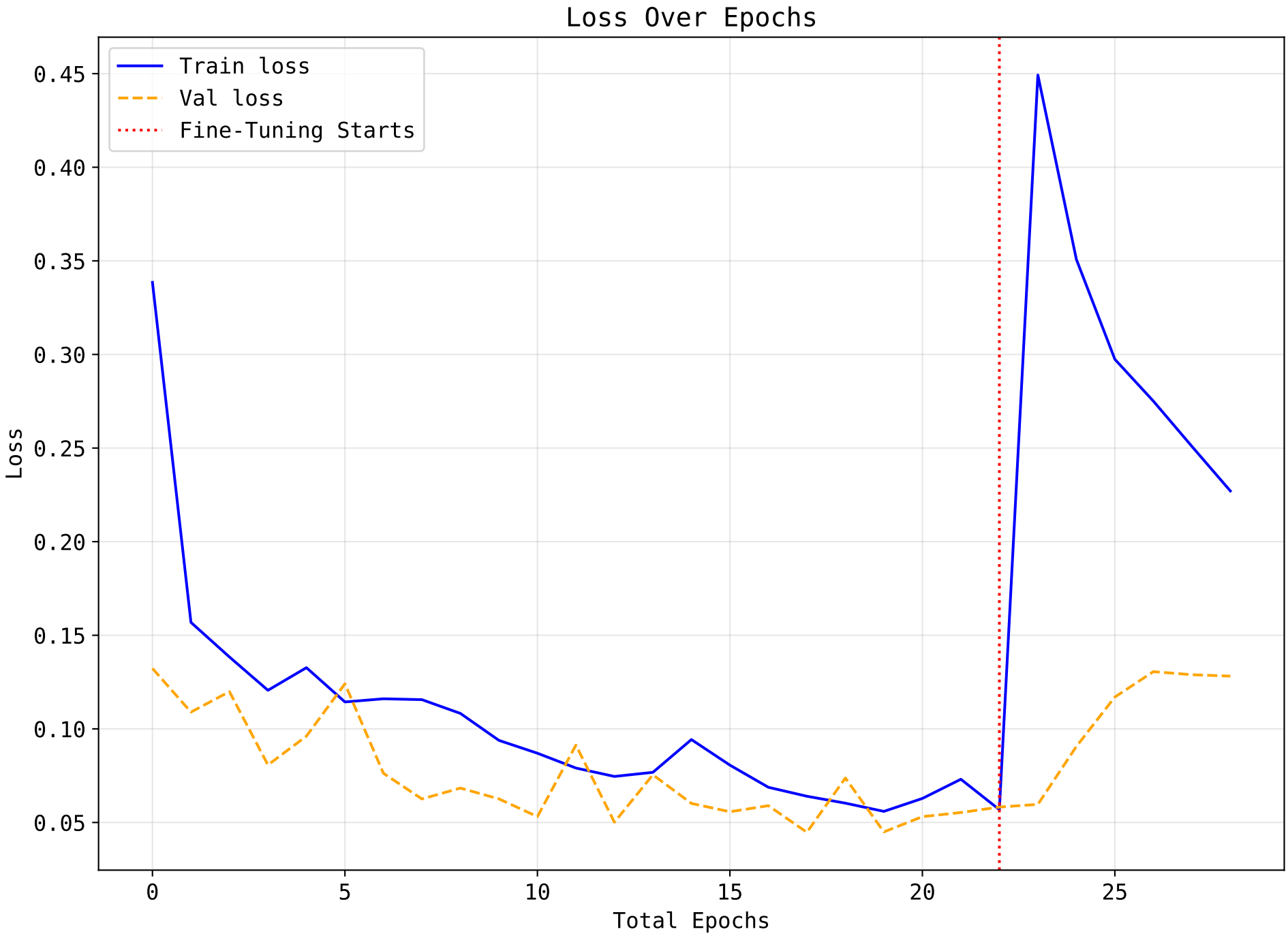
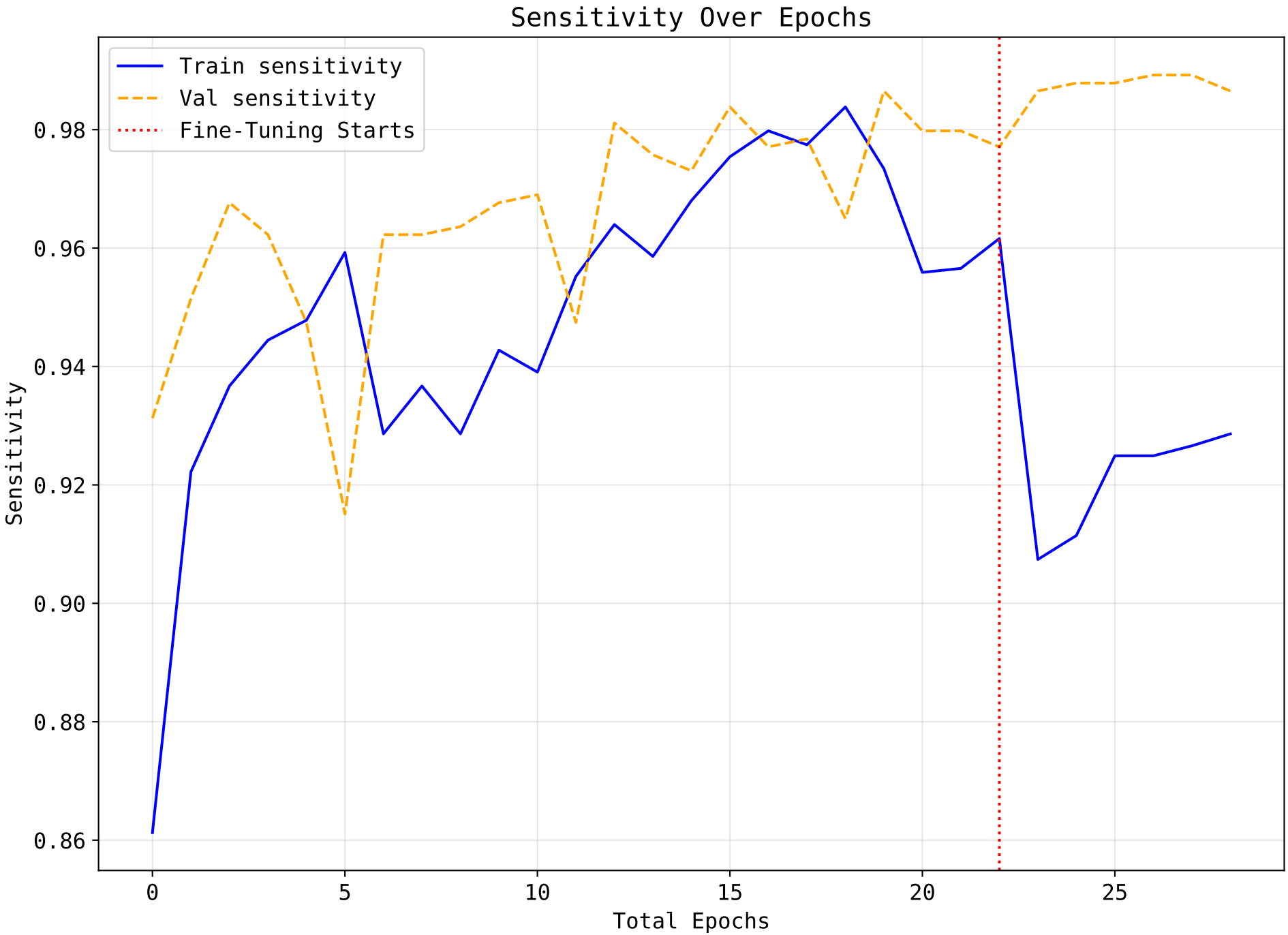
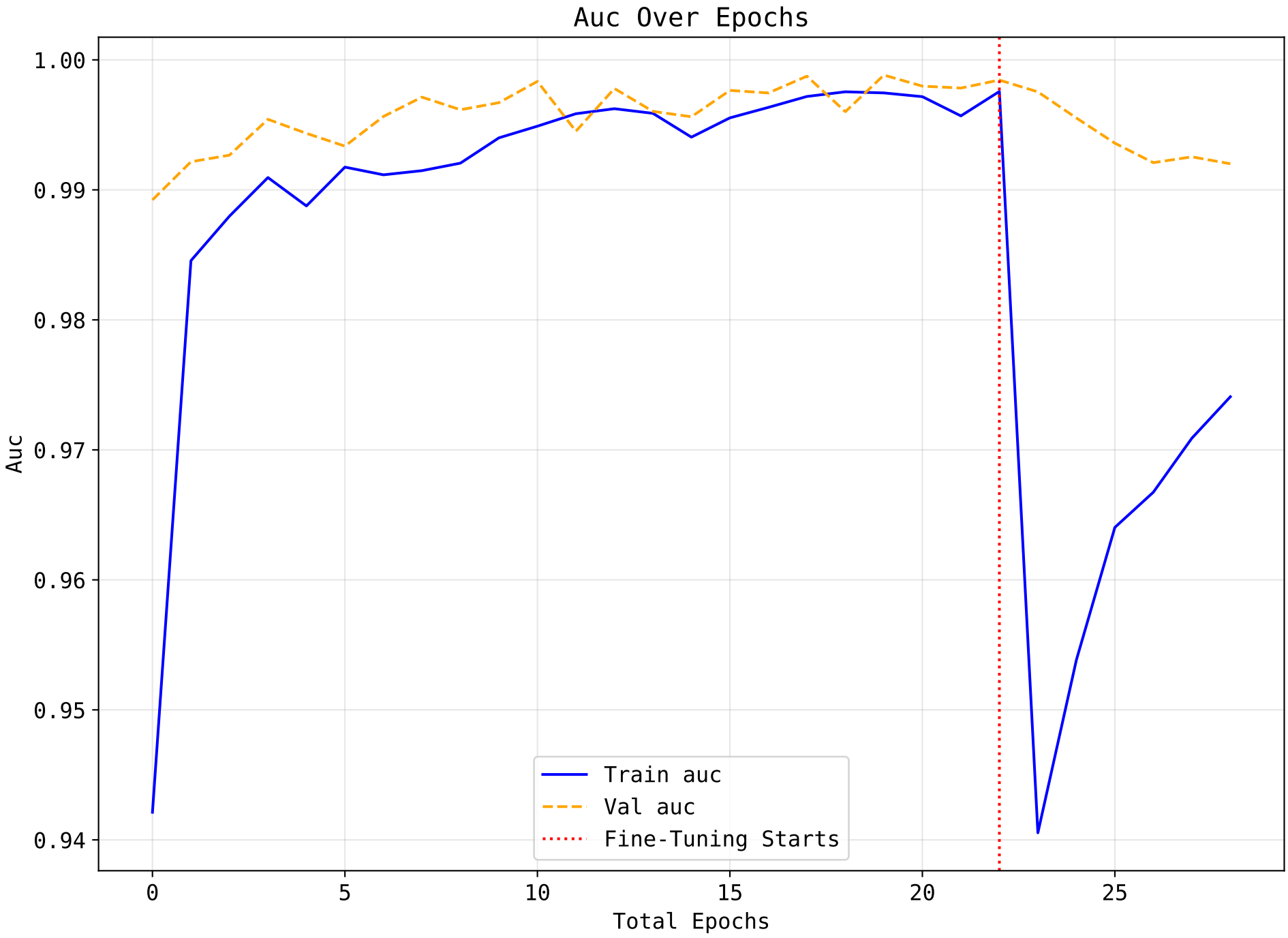
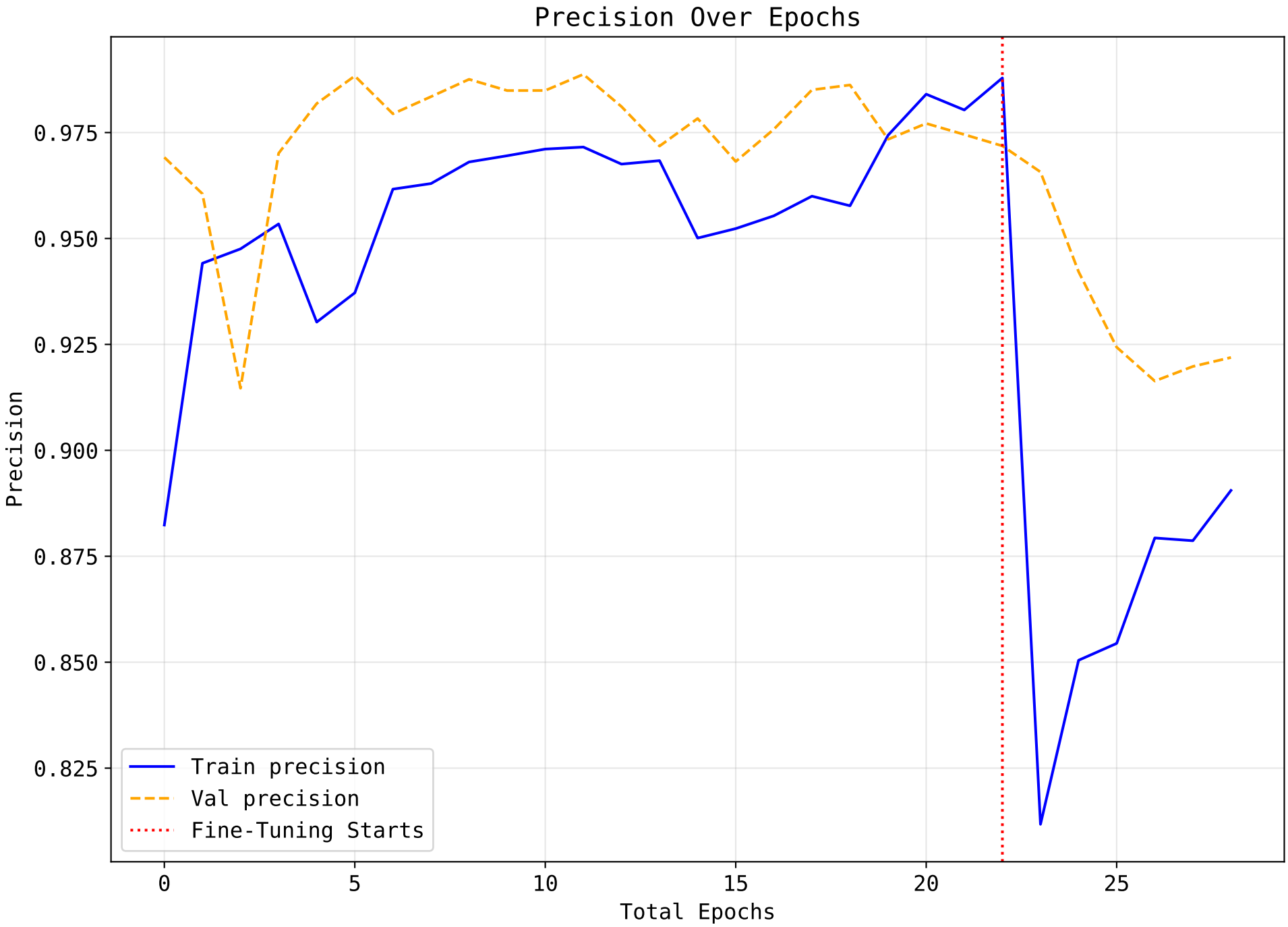
```
Model Evaluation Report
Started: 2026-04-12 23:10:24
Finished: 2026-04-12 23:21:24

Model Architecture: MobileNetV2
Early Stopping Patience: 5
Phase 1 Learning Rate: 0.01
Phase 2 Learning Rate: 1e-06
Phase 1 Epochs:           23 (Limit: 30)
Phase 2 Epochs:           6 (Limit: 50)

Test Loss                : 1.1686
Binary Accuracy          : 0.8289
Precision                : 0.7813
Sensitivity (Recall)     : 0.9054
AUC                      : 0.8847
Specificity: 0.7549

Optimal Threshold found: 0.9994
Optimised sensitivity: 0.8108
Optimised specificity:0.9281

Classification Report:
              precision    recall  f1-score   support

      Normal       0.89      0.75      0.82       306
    Cataract       0.78      0.91      0.84       296

    accuracy                           0.83       602
   macro avg       0.84      0.83      0.83       602
weighted avg       0.84      0.83      0.83       602
```

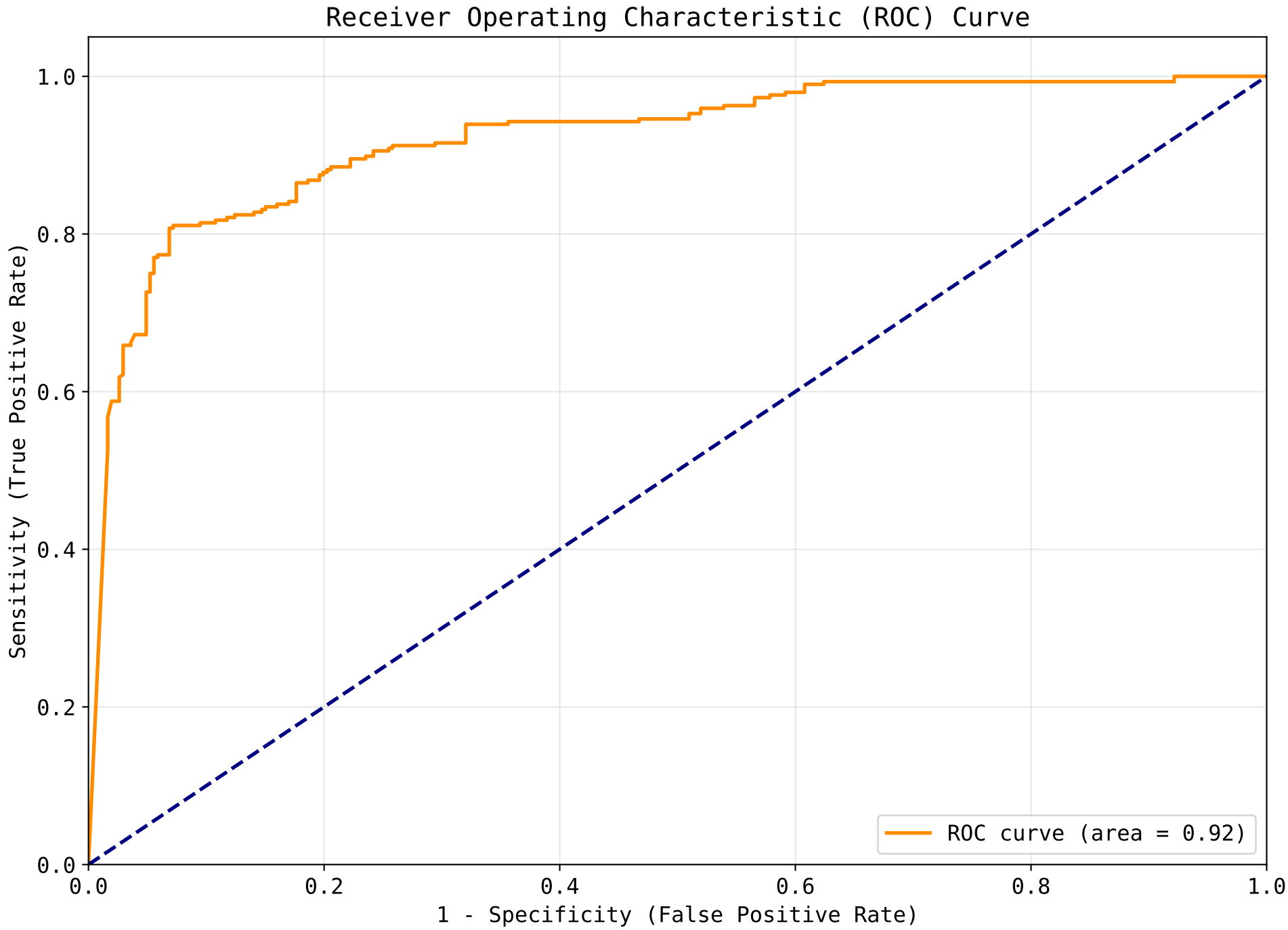

Accuracy Over Epochs
Accuracy
Total Epochs
0.98
0.96
0.94
0.92
0.90
0.88
0.86
0
5
10
15
20
25
Train accuracy
Val accuracy
Fine-Tuning Starts

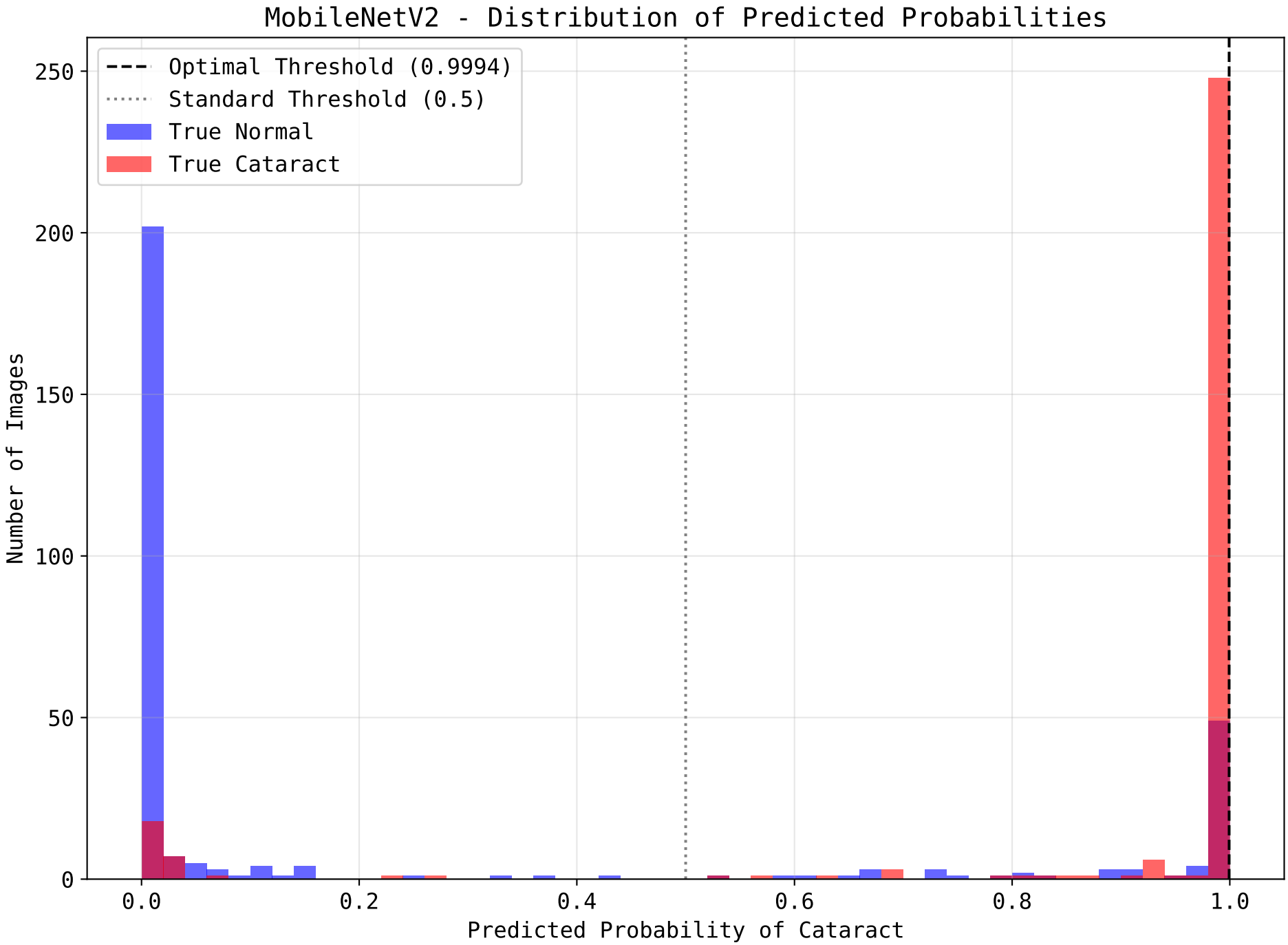

Loss Over Epochs
Train loss
Val loss
Fine-Tuning Starts
Loss
0.45
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0
5
10
15
20
25
Total Epochs

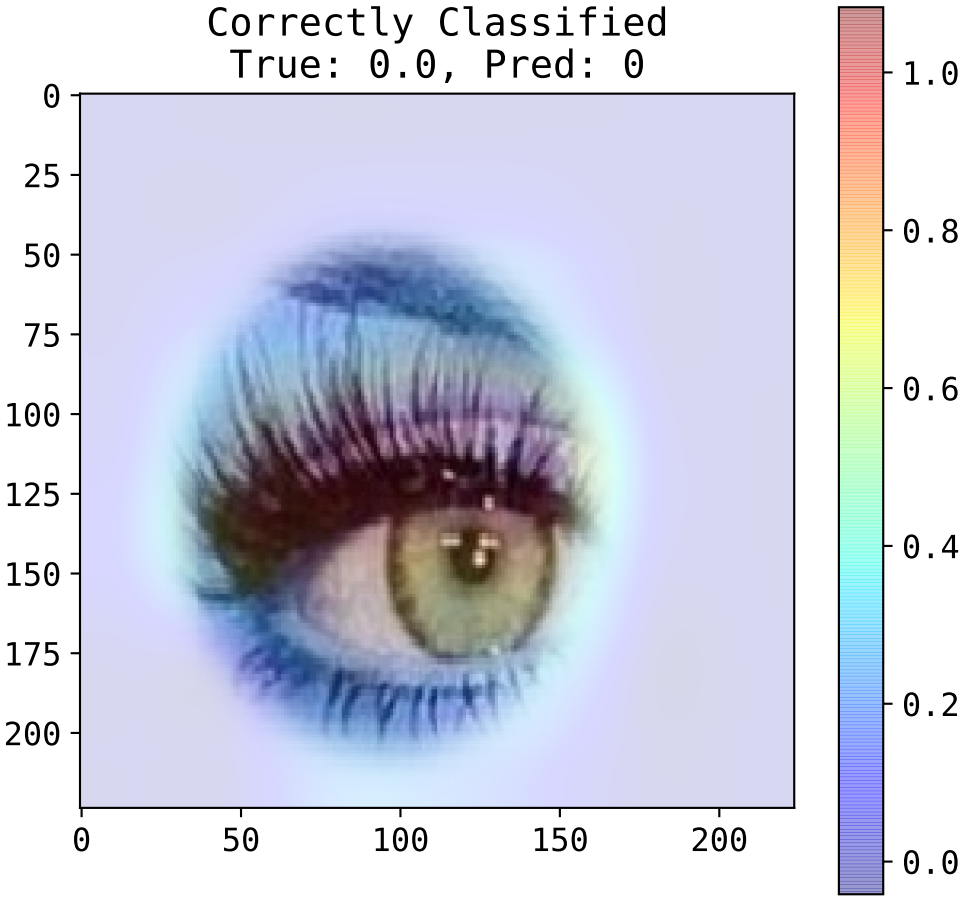

Sensitivity Over Epochs
Train sensitivity
Val sensitivity
Fine-Tuning Starts
Sensitivity
0.98
0.96
0.94
0.92
0.90
0.88
0.86
0
5
10
15
20
25
Total Epochs

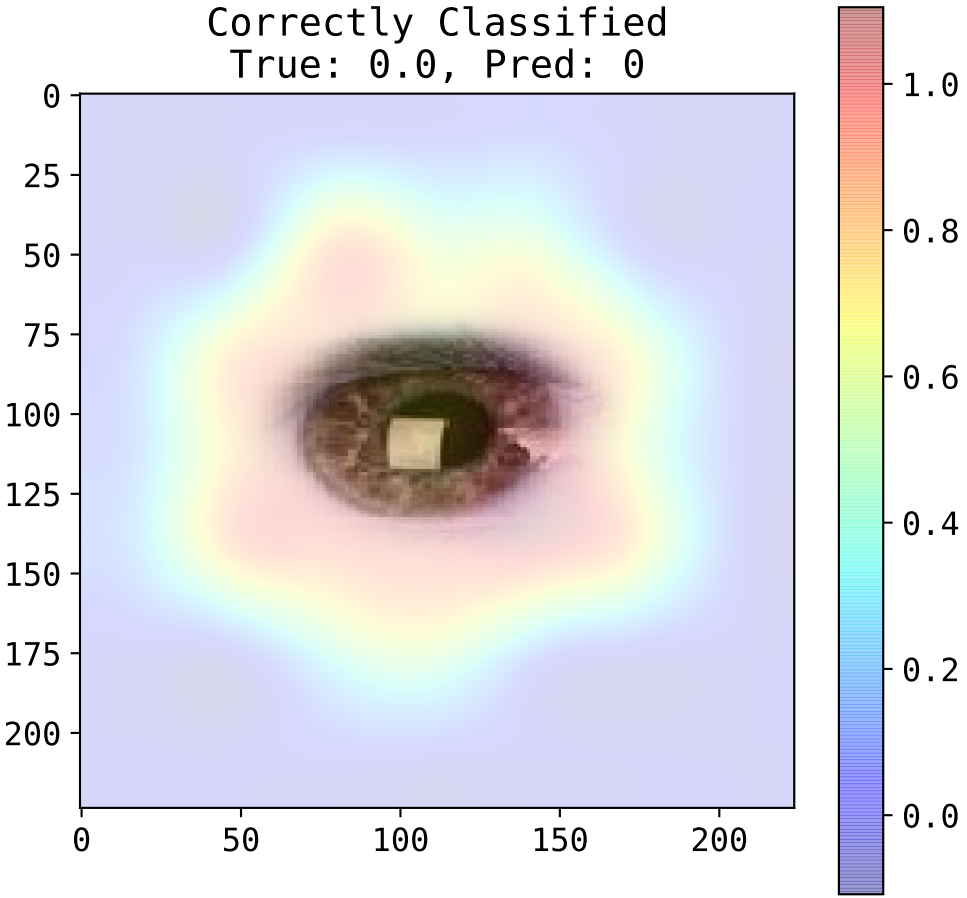

Auc Over Epochs
1.00
0.99
0.98
0.97
0.96
0.95
0.94
Auc
0
5
10
15
20
25
Total Epochs
Train auc
Val auc
Fine-Tuning Starts


Precision Over Epochs
Precision
0.975
0.950
0.925
0.900
0.875
0.850
0.825
0
5
10
15
20
25
Total Epochs
Train precision
Val precision
Fine-Tuning Starts


Receiver Operating Characteristic (ROC) Curve
Sensitivity (True Positive Rate)
1 - Specificity (False Positive Rate)
0.0
0.2
0.4
0.6
0.8
1.0
ROC curve (area = 0.92)


MobileNetV2 - Distribution of Predicted Probabilities
Optimal Threshold (0.9994)
Standard Threshold (0.5)
True Normal
True Cataract
Number of Images
Predicted Probability of Cataract
0
50
100
150
200
250
0.0
0.2
0.4
0.6
0.8
1.0


Correctly Classified
True: 0.0, Pred: 0
0
25
50
75
100
125
150
175
200
0
50
100
150
200
1.0
0.8
0.6
0.4
0.2
0.0


Correctly Classified
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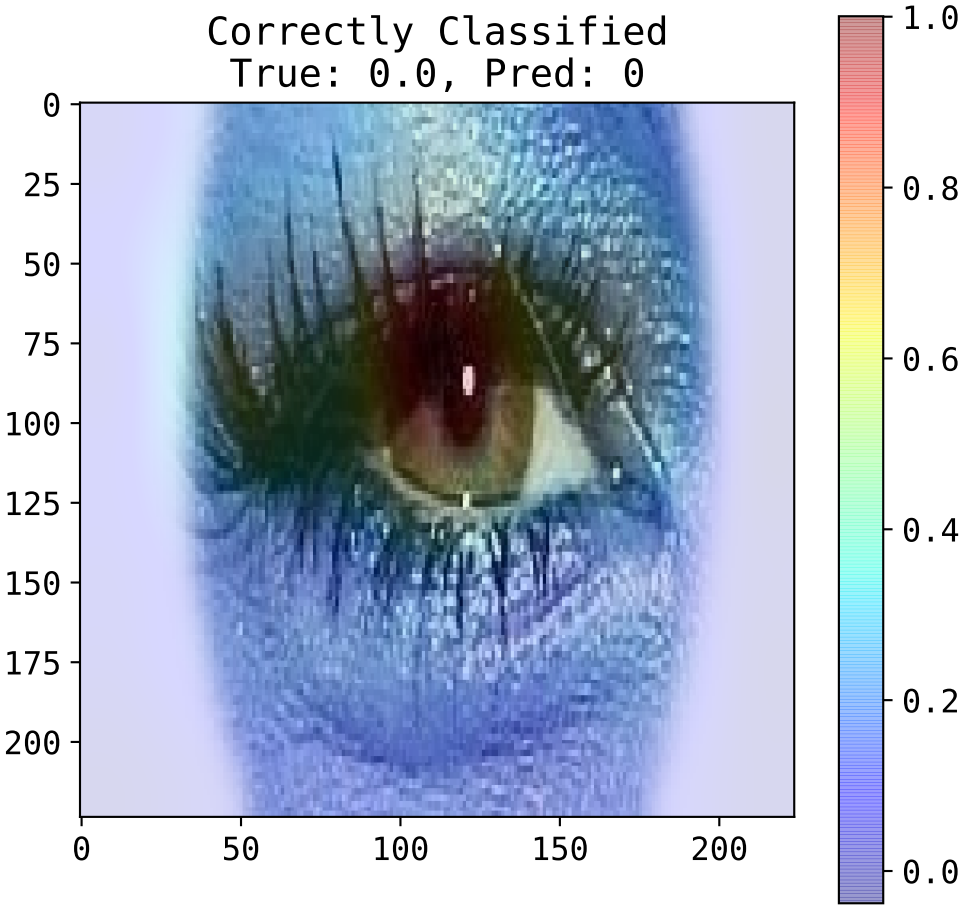
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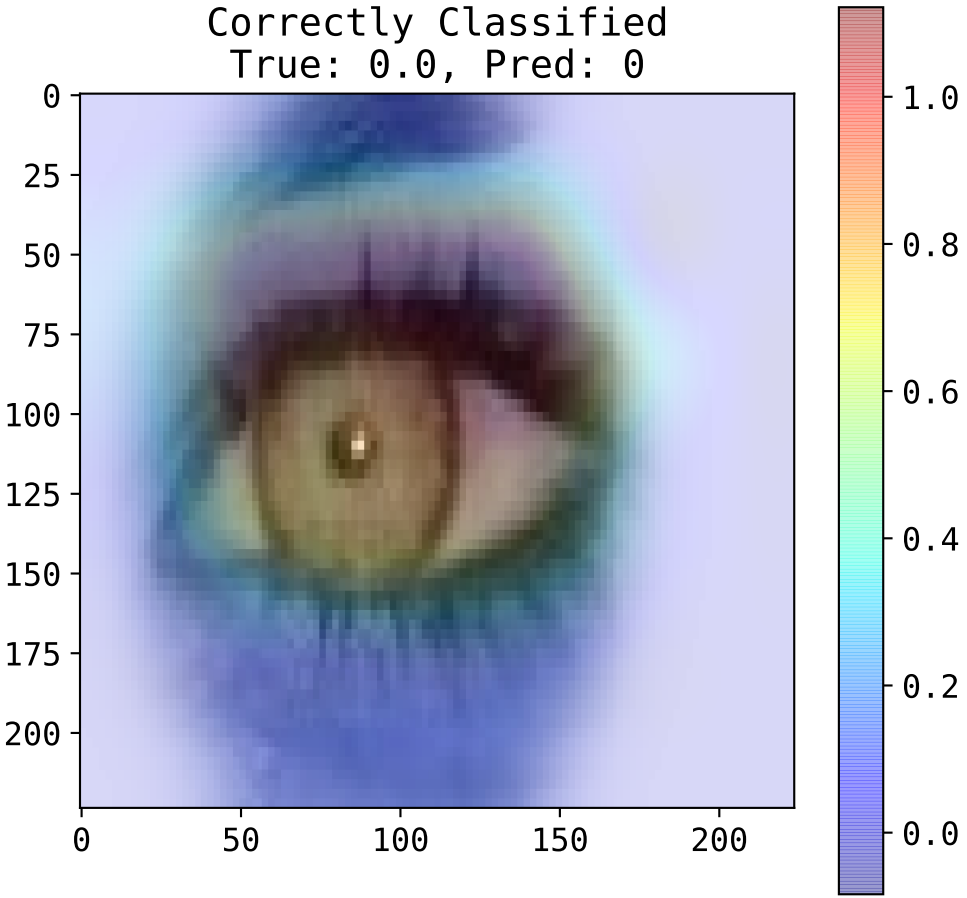
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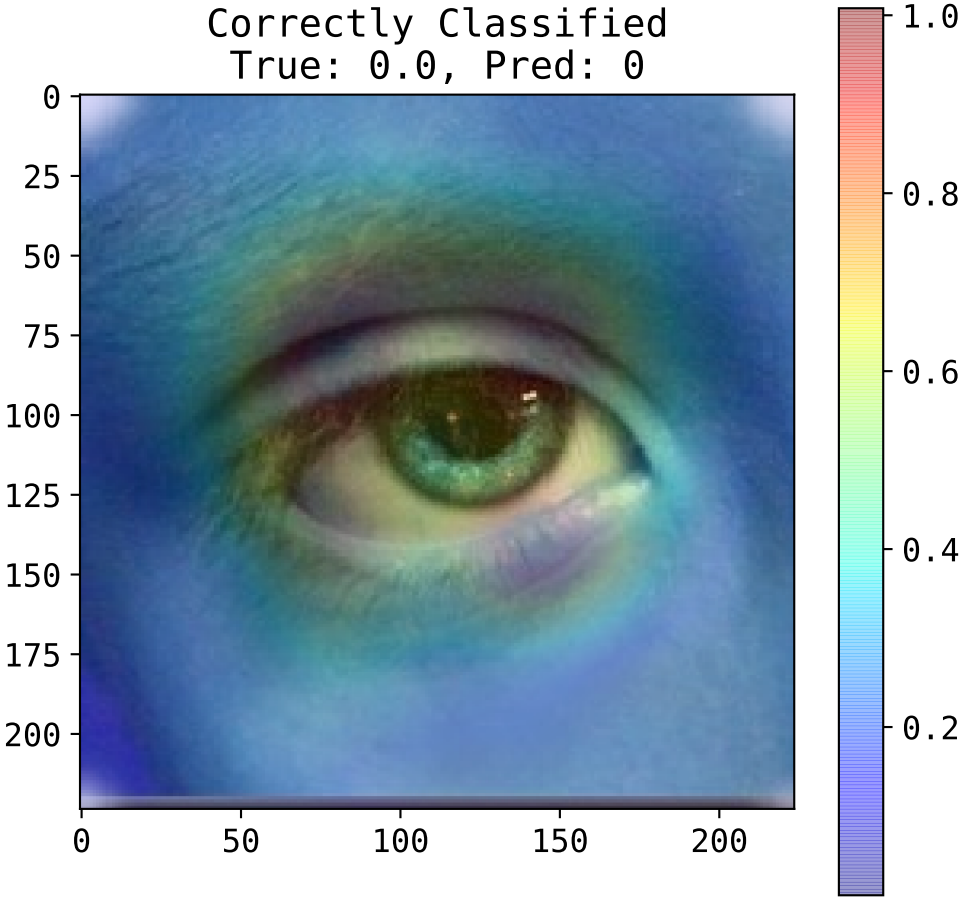
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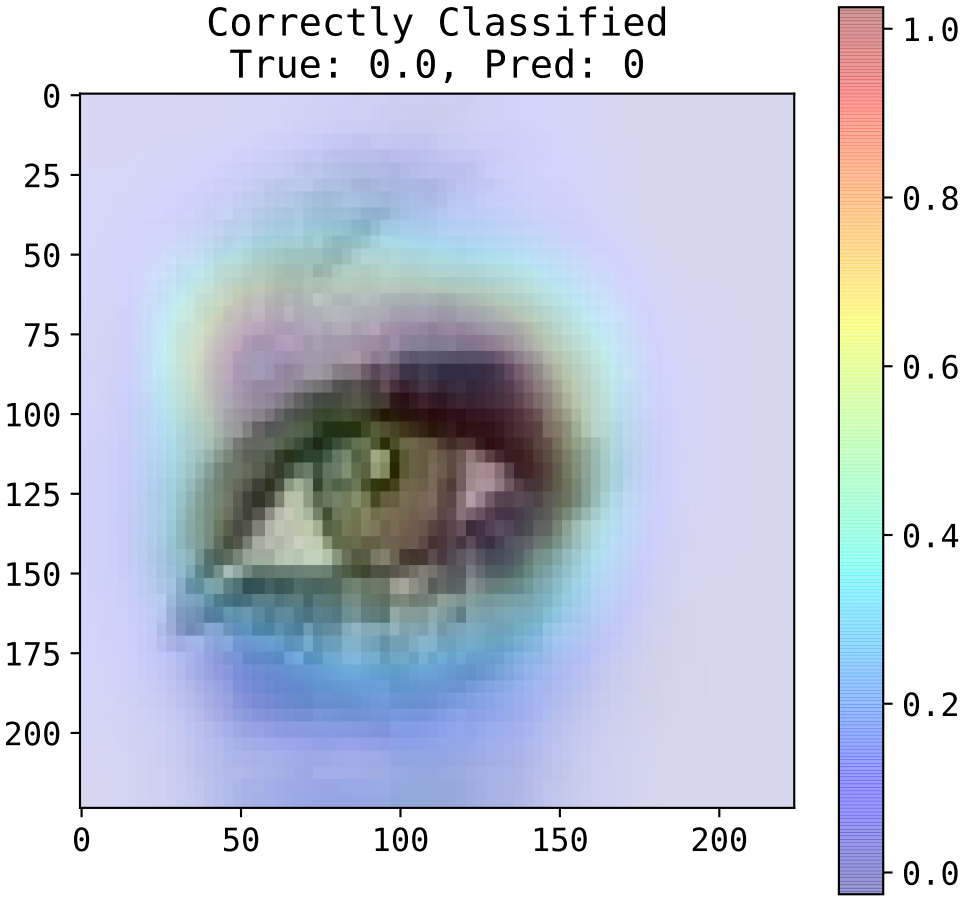
Correctly Classified
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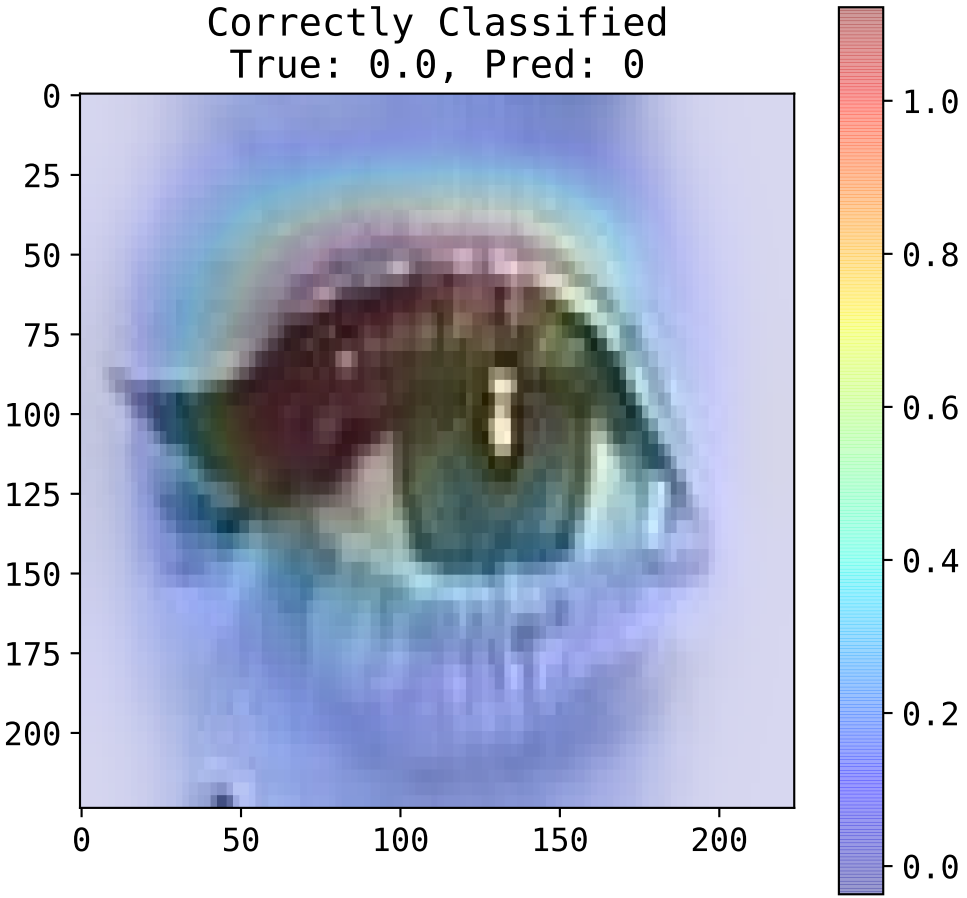
Correctly Classified
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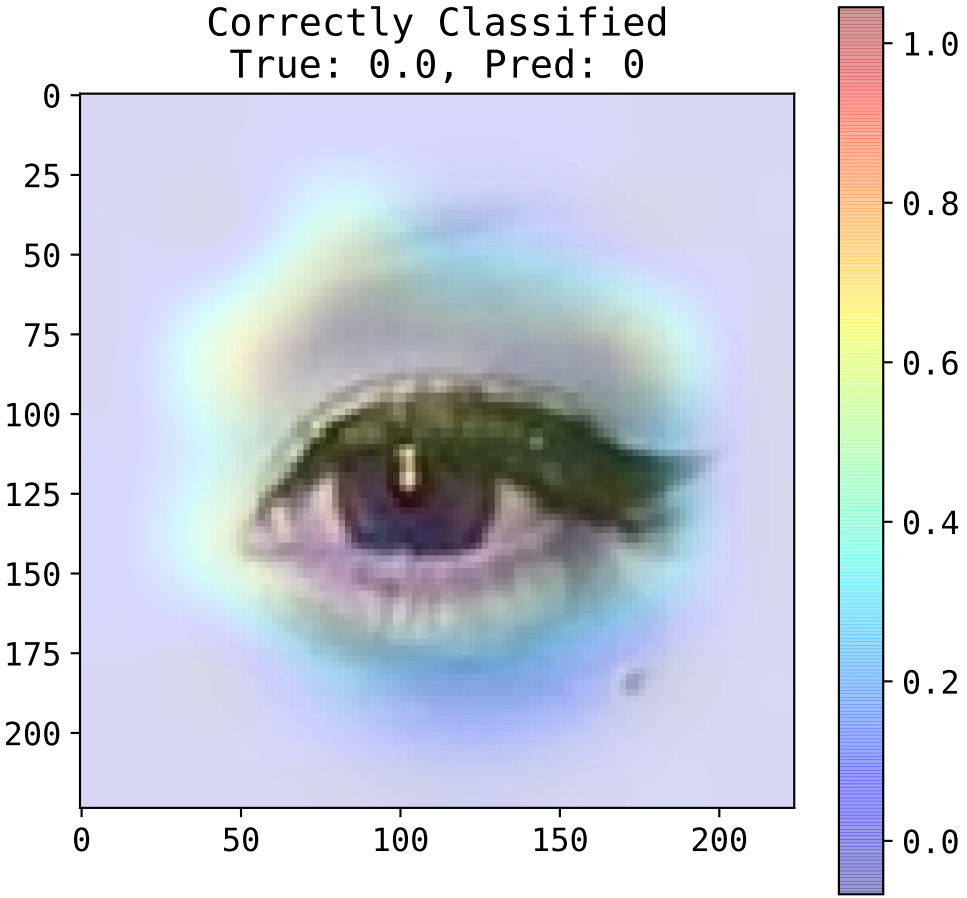
Correctly Classified
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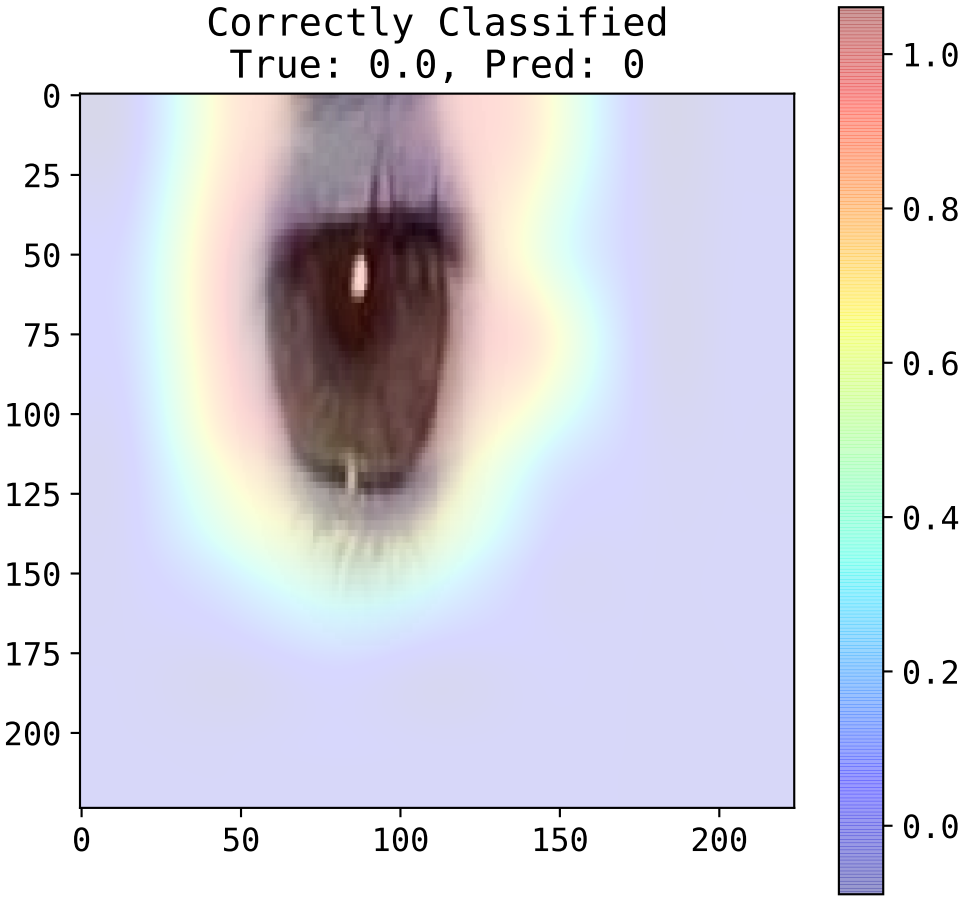
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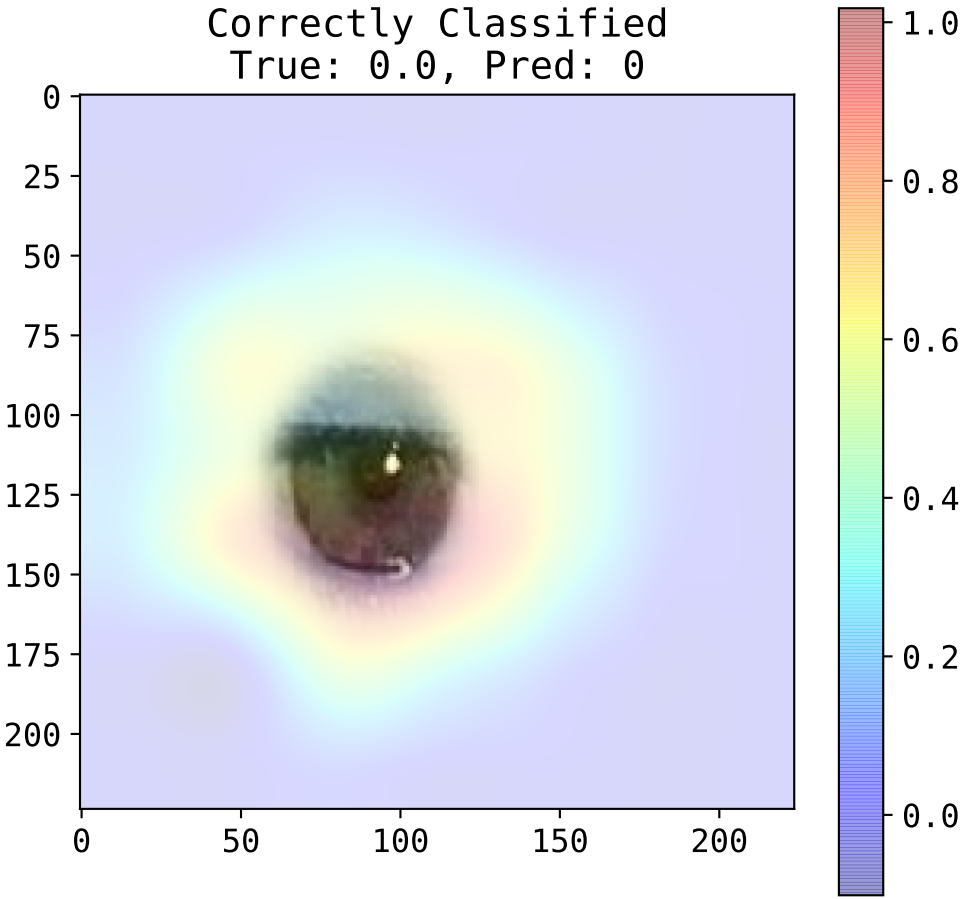
Correctly Classified
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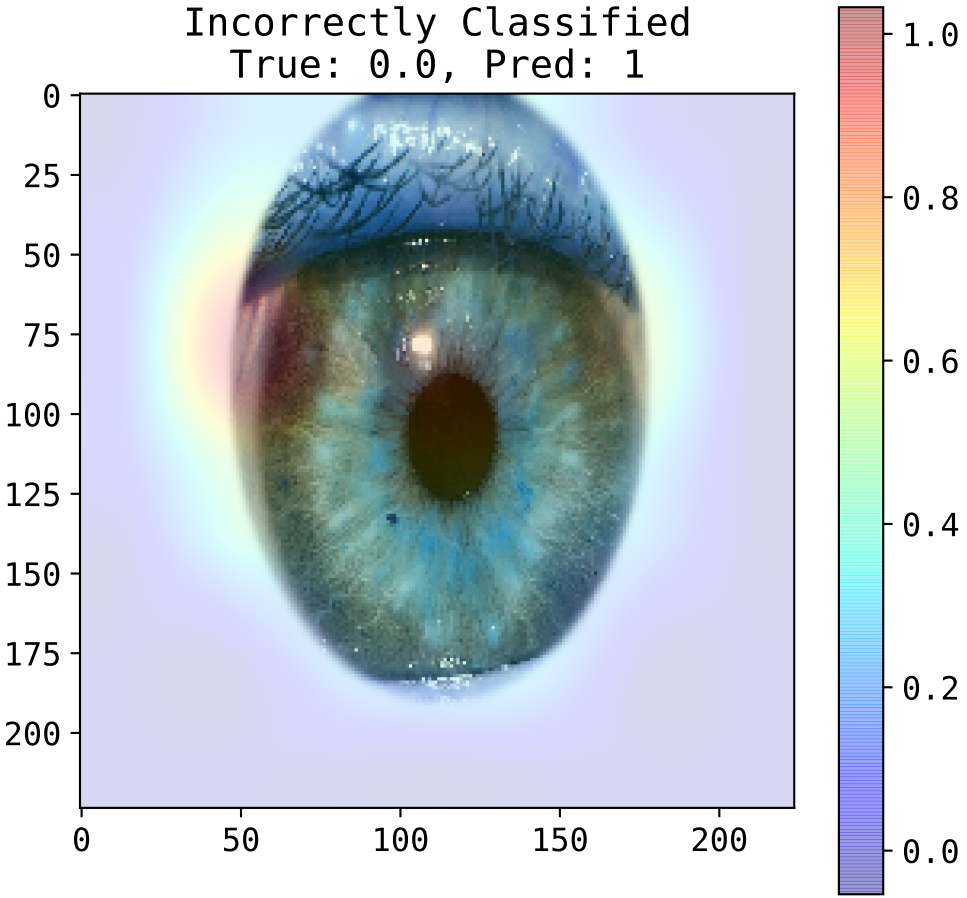
Incorrectly Classified
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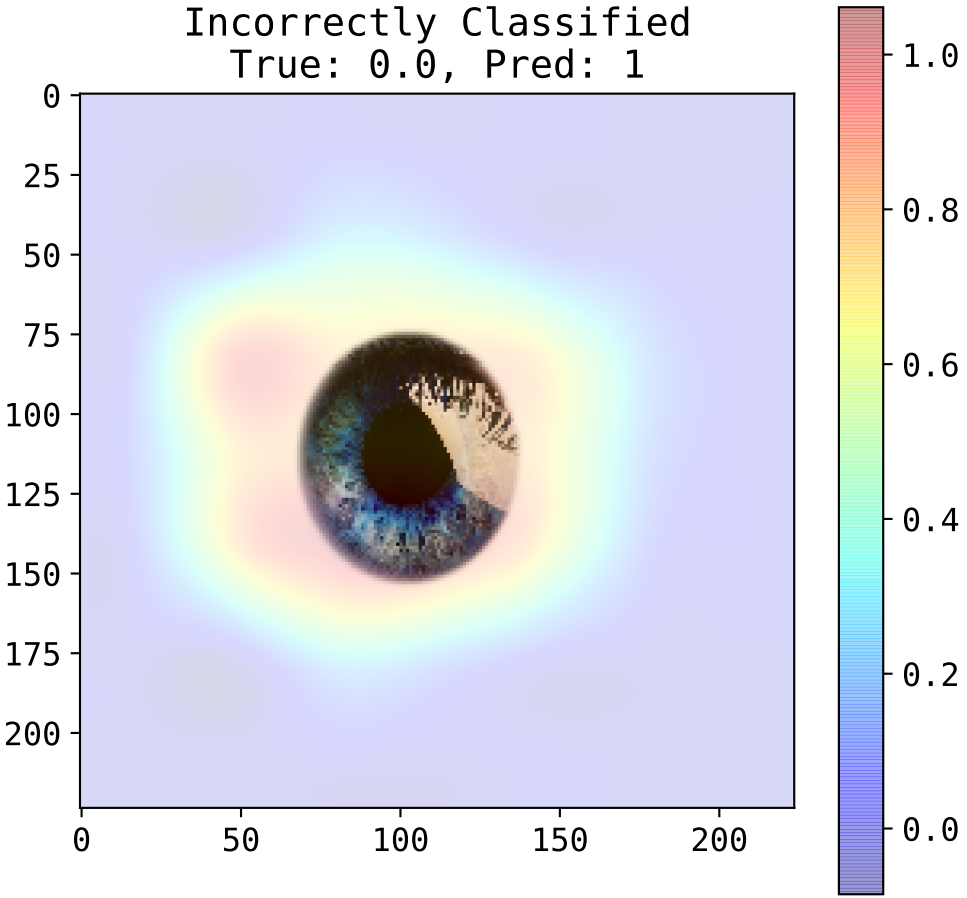
Incorrectly Classified
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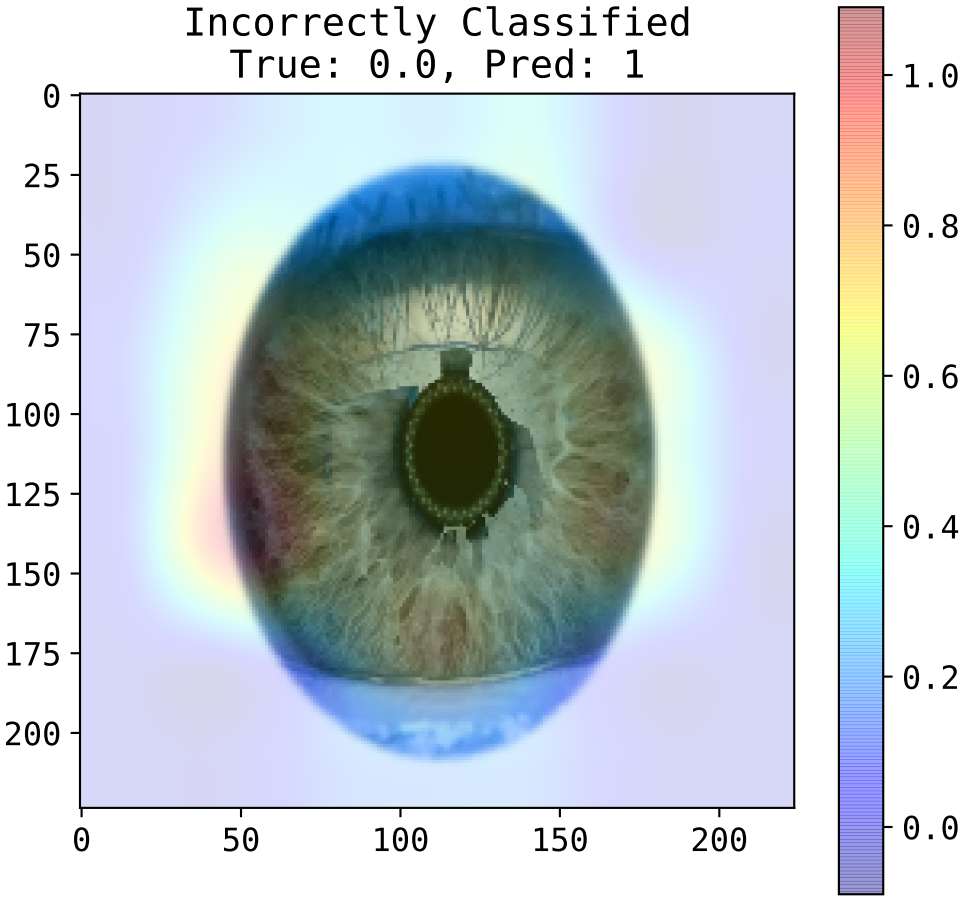
Incorrectly Classified
True: 0.0, Pred: 1
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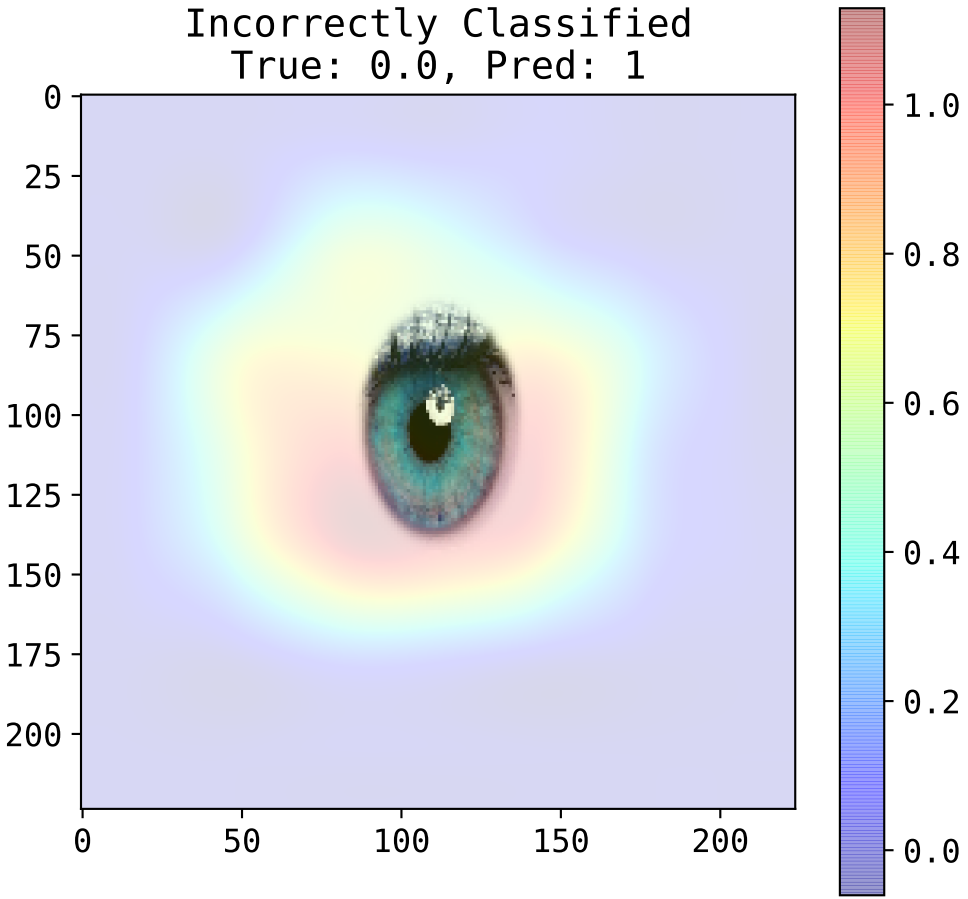
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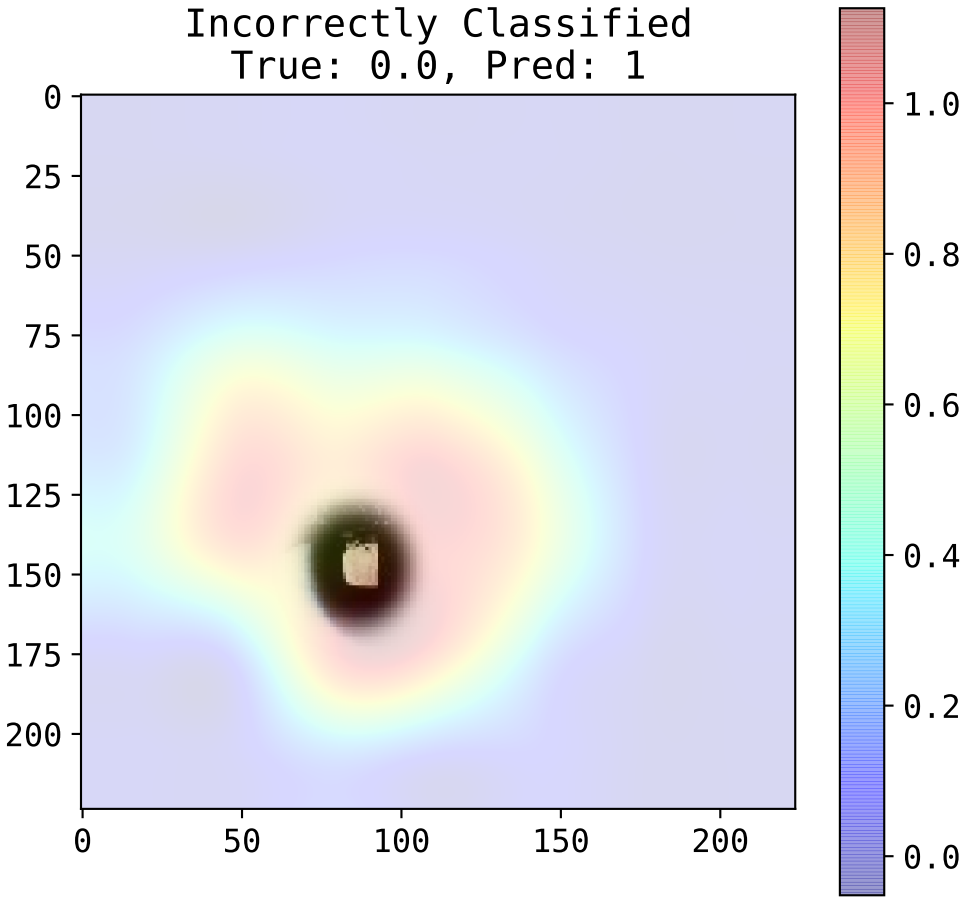
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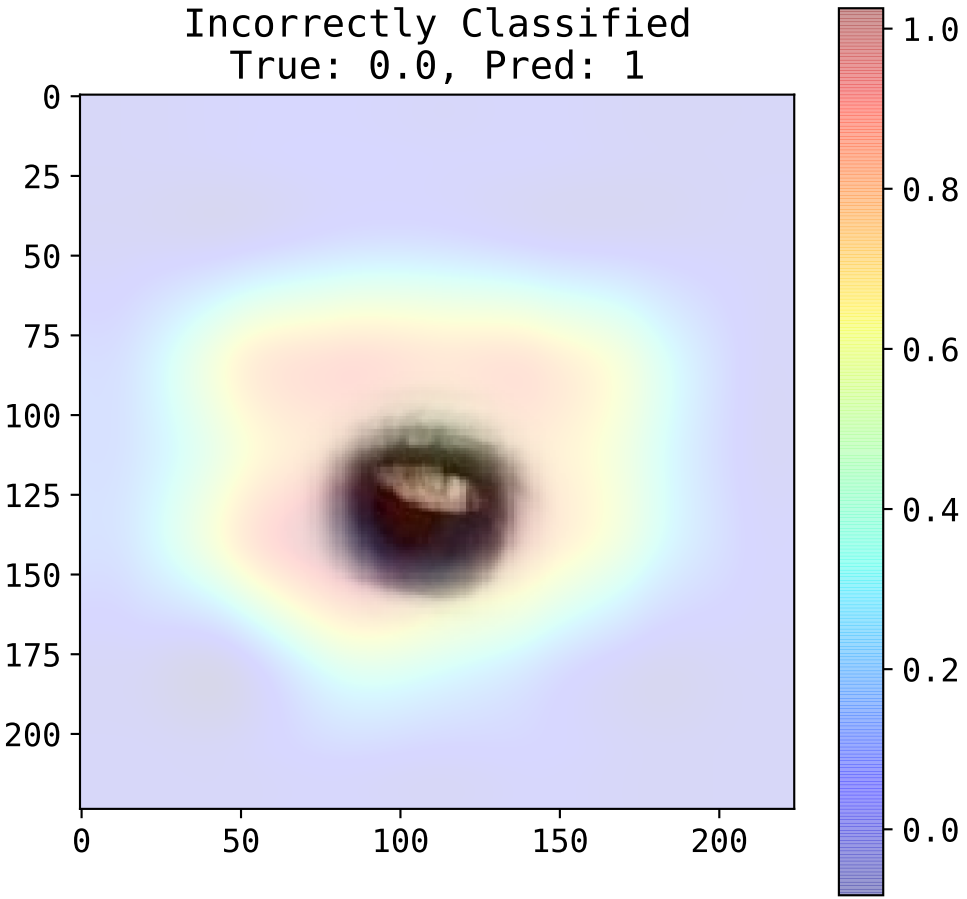
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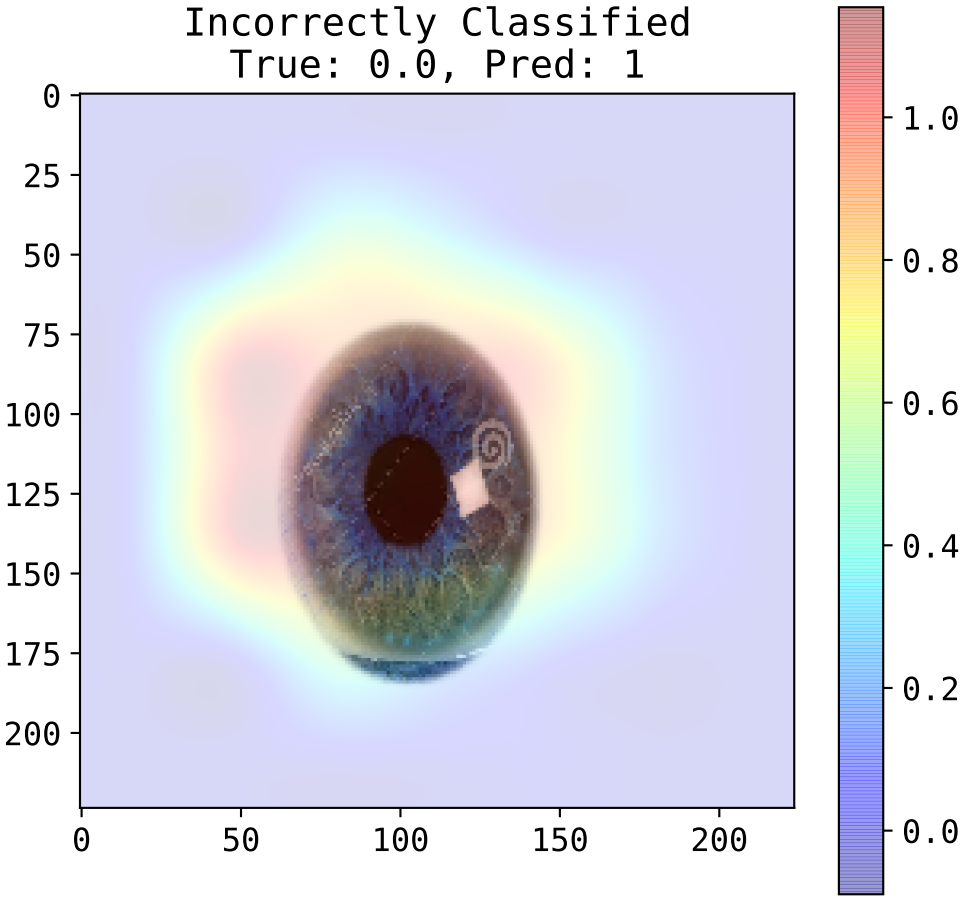
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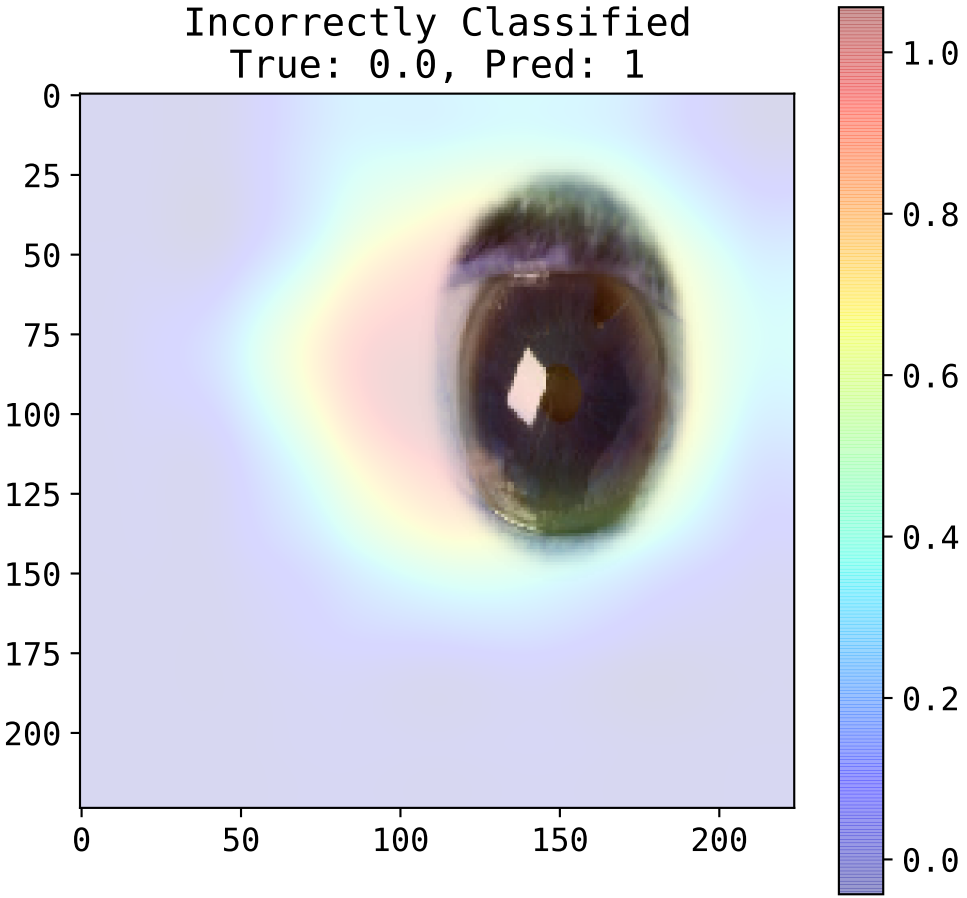
Incorrectly Classified
True: 0.0, Pred: 1
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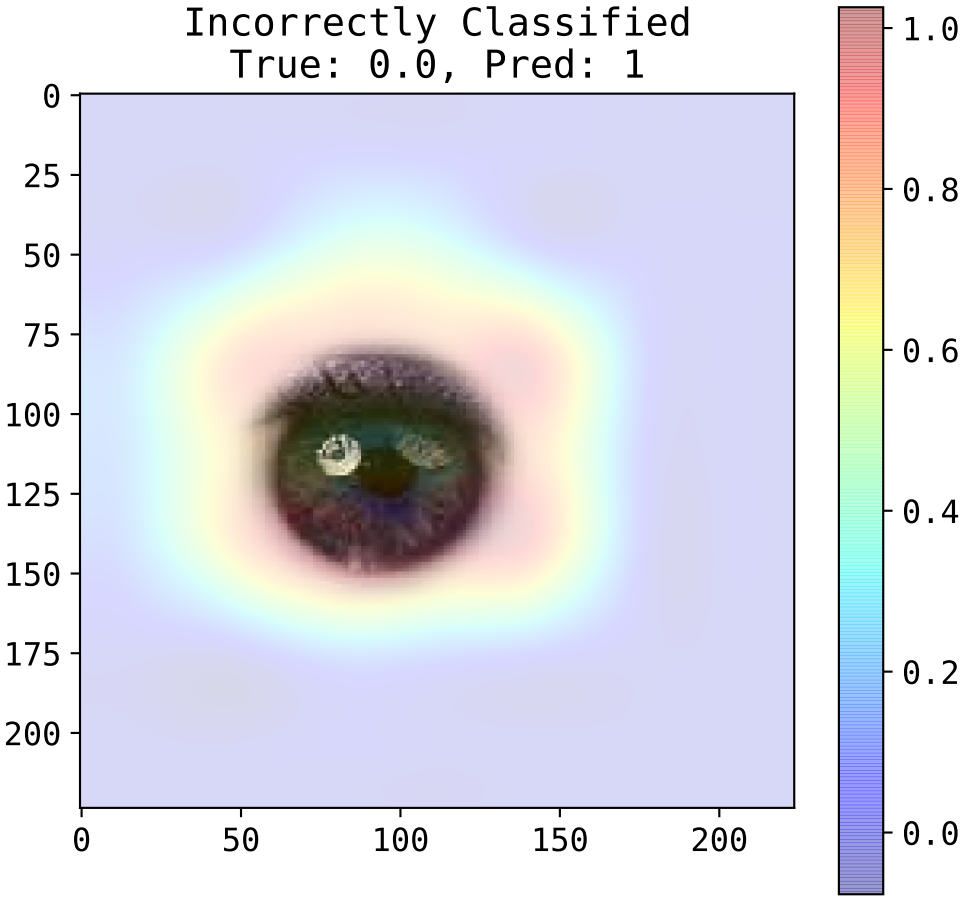
Incorrectly Classified
True: 0.0, Pred: 1
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1.0
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0.6
0.4
0.2
0.0